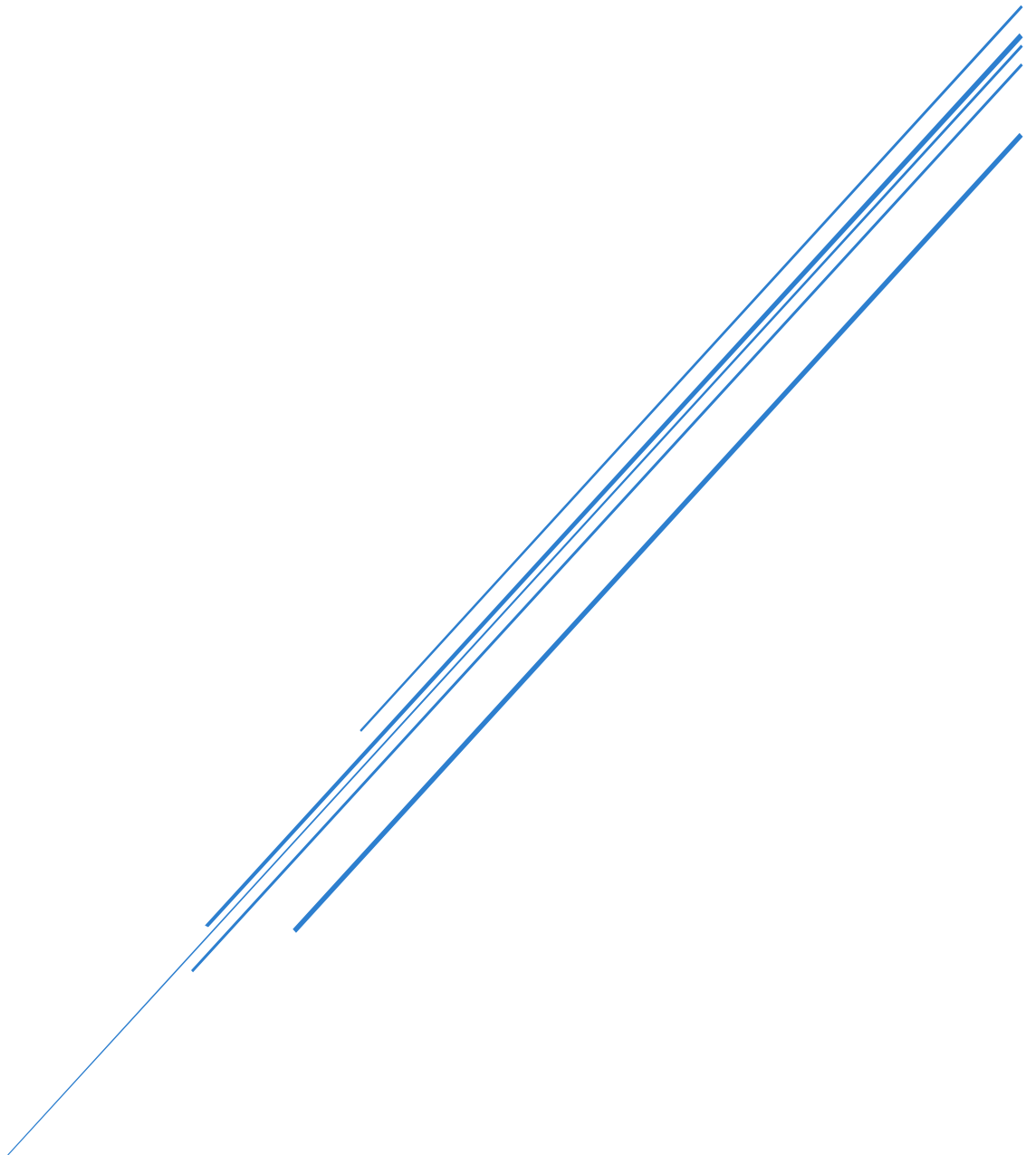


PRINCIPLES OF SECURITY: ASSIGNMENT 2

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Question 1

1. Introduction

In the context of Information and Communications Technology (ICT), Security means protecting the computer systems, network, software, and data against unauthorized access, attacks, damage, or theft (Paul Joore, 2021). Convenience is about how easy and efficient users can access and utilise technology with little to no hassle or time delay (BusinessMole, 2020).

At first glance it might be thought that both would improve simultaneously; however, Mark Ciampa (2024) suggests there is an inverse relationship between Security and Convenience. Simply put, as the level of security rises, so does the number of steps needed for each transaction, therefore reducing the overall efficiency and usability of the system. The added layers of security such as passwords, OTP's, Biometric scanning etc., make systems slower and less user friendly (Ciampa, 2024). This inverse relationship is evident within the South African Digital Banking Sector. Banks are constantly trying to find a balance between having sufficient security measures to prevent fraudulent transactions while at the same time ensuring they provide adequate levels of convenience to their customers (SABRIC, 2024).

2. Practical Example

South Africa's banking sector is a very good example of this trade off. Digital banking fraud has skyrocketed; has increased by 86%, from 52,000 cases in 2023 to nearly 98,000 for the year ending March 2024 with total estimated losses of R1.888 billion (TechAfrica , 2025). Banks are therefore implementing stronger and more robust security systems to combat this growing trend of criminal activity (SABRIC, 2024). Even though the additional security measures are effective at stopping unauthorised access into customer bank accounts; the added security measures will also create inconvenience when conducting banking activities.

Multi-Factor Authentication (MFA) is an excellent example of how Security and Convenience impact one another. Multi-Factor Authentication uses at least 2 different verification processes to confirm who you are before allowing you to gain access to your online banking account or perform any type of banking activity. Typically, MFA includes something you know (password/PIN), something you have (mobile device/OTP), and/or something you are (fingerprint/facial recognition) (Ciampa, 2024). While Multi-Factor Authentication is very successful at limiting unauthorised access into customer banking accounts (NCSC, 2024), it adds extra steps to the process. Therefore, banking becomes slower and more dependent upon the quality of the user's device(s) and connection speeds.

The way in which Standard Bank's Business Online platform has used two factor authentication (2FA) as a main method of securing accounts also illustrates this idea, and describes 2FA as "A security process in which the user provides two authentication factors to prove they are who they claim to be" and adds "An additional layer of security to prevent fraudsters from being able to steal logon credentials." (Standard Bank, n.d.). While the inclusion of 2FA does add a high degree of protection to the account, those clients with no mobile data coverage or no mobile phone signal at the time of a transaction will be unable

to obtain an in-app approval to enable them to authorise a payment. As such this represents one very good example of the inverse relationship mentioned earlier; because Standard Bank added so much more security using 2FA, the level of convenience for its customers was greatly reduced. Consequently, many customers rely on having connectivity available to have full access to their own money and, as we know, a lost signal can completely stop bank services.

Similarly, ABSA banking platform also uses numerous methods to protect each client's account. These include: 1) Login Protection that prevents a client's device from accessing passcodes, 2) requires Face or Touch ID to authenticate all logins and 3) uses 3D secure transaction approvals that redirect clients to the banking app where they must approve transactions through in-app approvals via SureCheck notifications (Absa, n.d.). In addition to these above-mentioned methods, if a client inputs their PIN or password wrongfully three consecutive times, the clients' online banking service will be disabled and they must either physically go to an ABSA branch or reset their credentials online (Absa, n.d.). As stated previously, every one of these measures creates a direct barrier between a client and their account. Therefore, every one of these measures also reduces the amount of convenience that is afforded by increasing the amount of security. For instance, while ABSA's online banking system may be highly secure as clients are limited to biometric only logins and mandatory in-app approvals when making transactions, there is a lot less convenience than would otherwise exist.

Capitec's strategy regarding security shows the inverse relationship very clearly. A user accesses the Capitec App and authorises transactions via a Remote PIN; if the user loses their device through theft, loss or breakage then the user will be required to complete an entirely new authentication process (Payments Afrika, 2014). Further, Capitec has recently implemented a four-hour delay for selfie verification (Capitec, 2025). This means that a legitimate customer who needs immediate access to their account after having lost their phone will have to wait up to four hours before being able to bank again. The users' feedback indicates that while the selfie verification requirement for one-off payments is considered by them to be a security feature, it is also described by them as "very frustrating" and complicates the payment process unnecessarily. Thus, Capitec has shown Ciampa's (2024) inverse relationship at work: as Capitec's security measures were enhanced from Remote PINS to selfie verification and the four-hour delay prior to re-enabling services, the ease of using these banking products was reduced in direct relation to the enhanced security measures. Users experienced longer transaction times and reduced availability in emergency situations.

Banking institutions across South Africa continue to enhance security measures in response to cybercrime risks, however the enhancement of security measures has resulted in banking becoming less convenient and more reliant upon technology.

3. Brief conclusion

In conclusion, it is safe to say that the banking sector in South Africa supports Ciampa (2024), who stated that "convenience" and "security" do not have a direct relationship. The banking industry has been adding layers of security, including multi-factor authentication, biometric identification, and approvals via apps, to fight back against growing cybercrime. These extra security steps make it more difficult, which makes the user experience more complicated and therefore reduces the convenience (SABRIC, 2024).

With each new layer of security, the customer will spend more time and effort to complete their transaction. This may decrease the level of convenience; however, the need for security outweighs the need for convenience to protect customers and financial institutions.

Question 2

- **Security Challenges and Concerns**

Biometric Authentication is a method of verifying identity through unique physical or behavioural characteristics such as fingerprint, face scan, iris scan or voice (Ciampa, 2024; IBM, 2024). Biometrics has been shown to provide greater levels of security compared to using passwords. Government secondary schools in South Africa may be unwilling to utilise biometrics based upon certain security risks.

The largest concern with Biometric Systems is the False Acceptance Rate (FAR): The percent of incorrect authentications by a Biometric System (Ciampa, 2024). In a school setting, this can result in trespassers or non-enrolled learners having the opportunity to enter school property when the biometric data of the unauthorised Individual is close enough to that of an enrolled student to allow entry into the system. High FAR's represent a serious failure of security; one that puts all learner's safety in jeopardy while they are at school. (Bayometric, n.d.).

Another Problematic aspect of Biometric Systems is the False Rejection Rate (FRR): The percent of incorrect denials of access to authorised individuals (Ciampa, 2024). With large numbers of students arriving simultaneously at entrances to government secondary schools throughout the country during peak hours; high FRR's will create massive backups at entrance gates; students who have arrived early may be marked absent and there will be massive disruptions to the daily schedule of the school (Bayometric, n.d.). To find a balance between the FAR and FRR, biometric systems measure a Crossover Error Rate (CER): The point at which both rates are equal. The lower the CER the more accurate & reliable the system. (Ciampa, 2024). However, low CER biometric systems are usually very expensive making them out of reach for most government secondary schools due to limited budgets.

One potential drawback to the use of biometrics in educational settings is the risk of biometric spoofing. Biometric spoofing occurs when an attacker utilises a false or duplicated version of a biometric trait to trick the system into recognising them (Ciampa, 2024). There are several types of biometric spoofing including: fingerprint spoofing - creating a copy of a fingerprint using silicone mould created from prints found in public places; face spoofing - displaying a photo or video of another person; iris spoofing - displaying a very high resolution printed version of another person's iris; voice spoofing - playing back recorded versions of other people's voices. Due to the ability of teenagers to create replicas of one another's fingerprints due to their familiarity with technology and each other and sharing items among themselves, the potential for biometric spoofing exists in many South African schools. An example of how easily a student can get their fingerprint stolen in a school setting is if they leave their fingerprint on a tabletop or smartphone screen and then has another student steal it by taking a mould of it and using the mould to spoof the original fingerprint. The student who stole the fingerprint can then log-in to whatever system requires biometric authentication that accepts fingerprints, such as logging in to access homework assignments online. (Oloid, 2026).

In addition to the technological issues outlined above, there are also other factors related specifically to South African government secondary schools that make it difficult for them to

implement biometrics. South Africa's Protection of Personal Information Act (POPIA, 2013) classifies biometric data as "special personal information," meaning there is a general prohibition on its processing unless strict legal conditions are met (POPIA, 2013). Since schools handle children under the age of eighteen years old, they will likely need to rely on exemptions based on reasonable necessity and/or consent. The fact that POPIA has provided for this type of protection for children makes it even more important that educational institutions comply with the provisions of POPIA, as children depend entirely on adults for making decisions regarding their own concerns (ENS, 2021).

Furthermore, a recent example illustrates the costs associated with failing to do so. A Polish school paid 4600 Euro (approximately R88,700) as a fine after processing students' fingerprints without having received permission (ENS, 2021). As stated earlier, schools have greater responsibilities than most organisations when it comes to complying with POPIA, as they collect and store personal information about individuals less than eighteen years of age. (Covington, 2024).

Many South African public schools do not have consistent electricity supply or internet connectivity needed for consistent use of biometric technology. Load-shedding alone will make systems unavailable during the peak times, causing children to be denied entry into their respective schools. Additionally, children's biological features change quickly during puberty, which means that as a child grows and their biometric data changes, it becomes increasingly difficult to match the learner with the previously enrolled biometric data. Additionally, when shared scanning devices are used there are hygienic issues involved and in an under-resourced school environment, the likelihood of vandalism or loss of equipment is a realistic concern.

● Practical examples

There are several potential forms of biometrics that can be used within South African secondary schools. Each has practical applications and possible risks related to security.

Fingerprint recognition - This method involves students scanning their fingerprints upon entering the school building for attendance. The advantages of this form of recognition include ease of use and low cost. Disadvantages include problems with damaged or dirty fingers causing higher False Rejection Rate (FRR). Fingers can also be duplicated and used fraudulently to spoof an individual's fingerprint. (Ciampa, 2024; IBM, 2024).

Facial recognition - This system allows for automatic identification of individuals through face recognition using cameras installed at entrances to buildings. The advantage includes increased convenience. The disadvantages include being affected by lighting and changes in a student's facial characteristics due to age. Face recognition can also be spoofed with photographs or video. (Ciampa, 2024; IBM, 2024).

Voice recognition - Requires students to state their name or other information to validate their identity. A disadvantage of this system is decreased accuracy in noisy environments such as classrooms and schools. In addition, since students' voices change during puberty, there will likely be an increase in False Rejection Rates (FRR) (Ciampa, 2024; IBM, 2024).

Iris recognition - This system requires a student to look directly into a camera/scanner that reads the unique patterns in the iris to identify them. Iris recognition is one of the most secure methods available because of its high accuracy rate and difficulty in duplication.

However, this method is extremely expensive and not feasible for many government funded schools. Some additional disadvantages of this method include problems caused by the need for students to remove their glasses prior to scanning, eye conditions, or positioning difficulties when attempting to place their eyes on the correct spot to read their iris. These factors contribute to increases in False Rejection Rates (FRR). (Ciampa, 2024; IBM, 2024).

In summary, although biometric systems provide better security than traditional systems, they create new vulnerabilities including errors and spoofing that must be considered by the schools.

Question 3

1. Definitions and Differences

The Domain Name System (DNS) is basically the worldwide web's telephone book as it converts human readable domain names such as "www.fnb.co.za" into the numerical Internet Protocol (IP) addresses that computer systems are able to understand so they can find and connect to an appropriate website (Cloudflare, n.d.). Therefore, when you type in a URL from your desktop or mobile device, your system will send out a request to a DNS server asking for the proper IP address; once received by your system, the web browser then makes a connection to the requested website. Since the original designers of DNS did not have any verification processes built into its design principles of trust, it has been targeted by attackers (Cloudflare, n.d.; Ciampa, 2024).

DNS Poisoning (Cache Poisoning):

DNS Cache Poisoning is the act of putting false data into a DNS Cache so when you run a DNS Query it will give you an incorrect response and send you to the wrong site (Cloudflare, n.d.). The attackers take advantage of the fact that most DNS Servers temporarily store, or 'cache', the results of previous lookups to help speed up DNS resolving (Ciampa, 2024). An attacker injects forged DNS records into this cache by impersonating a legitimate DNS nameserver and sends the fake record to the DNS resolver before it receives the real record. Most times, there is no way to validate the data in a caches so the false data remains in the cache until it expires with the Time To Live (TTL) or is manually removed (Cloudflare, n.d.). Resulting in all the users that use the same DNS server to be redirected to the fake website without any indication of a problem. The cache poisoning attack targets the DNS server's cache, not a device directly, therefore one poisoned server can affect thousands of users at once.

One of the best examples of cache poisoning in the real world occurred in 2008. Dan Kaminsky, a security researcher, identified a fundamental flaw in the DNS protocol that would enable most nameservers to be easily vulnerable to cache poisoning attacks which would enable many types of attacks such as impersonation of websites, interception of email and bypassing authentication (eSecurity Planet, 2016; Wikipedia, 2026). Kaminsky immediately contacted The Department of Homeland Security and executives at Cisco and Microsoft to co-ordinately work on fixing the Issue, which was released on July 8th, 2008, (Wikipedia, 2026). Although the Kaminsky attack was able to show how damaging DNS cache poisoning can be at scale, it also helped accelerate the adoption of using DNSSEC as a defending standard (eSecurity Planet, 2016).

DNS Hijacking:

DNS Hijacking, also known as DNS Redirection, is an entirely different type of attack from the one discussed above. While DNS poisoning focused on inserting false data into a caching system, in DNS hijacking, the attacker has complete access to the DNS resolution process itself. To achieve this, the attacker manipulates the DNS settings, typically by infecting the victim's computer with malware and then taking control of their routers, intercepting DNS traffic, and/or hacking into the communication flow of the DNS service (Kaspersky, n.d.). There are three primary ways for attackers to carry out such manipulation:

local DNS hijacking is possible when a user installs Trojan malware on his/her own computer to create new local DNS configurations;
router DNS hijacking occurs when an attacker uses weak passwords or exploits a vulnerability in the firmware of a router to change its DNS configuration so that all devices connected to that network will use the altered DNS configuration;
rogue DNS server attacks occur when a malicious party compromises a legitimate DNS server, thereby modifying the DNS records on that server permanently (Kaspersky, n.d.; Ciampa, 2024).

When a DNS hijacking attack occurs, the attacker redirects DNS query requests to a different domain name server by changing DNS settings, allowing them to control how domain names are resolved and permanently direct users to malicious or fake websites. The key difference is that DNS poisoning corrupts cached DNS data temporarily, while DNS hijacking alters actual DNS settings or records, giving attackers longer-term control over how DNS queries are resolved (Cloudflare, n.d.).

A recent example of a large-scale DNS hijacking was carried out against internet companies including telecoms and governments in the Middle East, Europe, North Africa and North America. Attackers used previously compromised log-in credentials to point users to fake versions of popular web sites where they could enter their login credentials and other sensitive information. Cloudflare states that they were able to track the source of these attacks back to IP addresses based in Iran (Cloudflare, n.d.).

Key Differences between DNS Poisoning and DNS Hijacking:

Although both types of attacks use DNS systems to divert users to unintended destinations and cannot be detected by most users, there are many differences in how each type of attack works (Ciampa, 2024; Cloudflare, n.d.):

Feature	DNS Poisoning	DNS Hijacking
What is attacked	DNS cache memory	DNS settings/infrastructure
Method	Injecting forged records into cache	Taking direct control of DNS settings via malware, router access, or server compromise
Access required	No direct device access needed	Requires access to device, router, or DNS server
Scale	Can affect thousands of users via one server	Ranges from a single device to entire networks
Duration	Until TTL expires or cache is cleared	Persistent until attacker is removed
Primary goal	Redirect users to fake sites	Redirect, intercept, spy on, or censor traffic

2. Intended Harm

DNS poisoning and DNS hijacking are two different types of attacks using the trust users place in the DNS system. However, both types of attacks create harm in different ways (Ciampa, 2024).

Harm from DNS Poisoning:

In general, the objective of DNS poisoning is to send unsuspecting users to malicious sites that look like real sites. When an individual user receives a poisoned DNS response, they will receive a phishing attack. Phishing attacks allow criminals to use a fake version of their bank, retailer, or government site to obtain an individual's username/password, account number, and financial information (Cloudflare, n.d.). As stated earlier, a poisoned DNS cache can redirect users to malicious or fake websites without their knowledge. This occurs when individuals believe they are logging into a safe site but are instead entering their information into a duplicate of the site being run by an attacker. Malicious redirects are often used to facilitate phishing, malware distribution, and/or large-scale data breaches (Comparitech, 2025). Organisations are at risk of experiencing reputational damage as well as losing their customers' trust. Organisations may also experience significant financial losses due to the cost of fraud committed against them. Additionally, organisations who have customer data stolen may face regulatory action such as fines or lawsuits (Ciampa, 2024). Since one poisoned DNS server can affect every user using it for query purposes, the amount of harm caused can be massive.

Harm from DNS Hijacking:

DNS Hijacking can be more damaging due to its intended nature and the persistence of the damage. The effects on a single user include an ability for the hacker to initiate a phishing scam; this includes presenting the victim with what appears to be a legitimate version of a website however it leads to stealing of credentials such as passwords, login credentials and credit card numbers. For a customer facing business website, the effects would allow hackers to reroute web page visitors to fake webpages (Kaspersky, n.d.). Furthermore beyond the direct loss of credentials through the theft of the above mentioned items, DNS hijacking allows hackers to execute man-in-the-middle attacks which enable them to intercept and read all traffic between a user and a server; neither the user nor the server will be aware of this interception (Kaspersky, n.d.; Imperva, n.d.). A number of Governments have utilised DNS hijacking for censorship purposes, by routing users to Government approved sites, whilst many Internet Service Providers have also used DNS hijacking for statistical collection purposes and returning advertisements to users who visit unknown domain names (Imperva, n.d.). When an organisation is subjected to a successful DNS hijacking attack there could be extended periods of time before they become aware that their systems have been compromised allowing the hackers to potentially remain undetected for months or longer and harvest various types of credentials and sensitive communications from these compromised systems as was clearly illustrated in the Global DNS hijacking campaign targeting Government entities across multiple countries (Cloudflare, n.d.).

3. Effectiveness of DNS Poisoning

Although Cybersecurity has made significant advancements since the original development of DNS poisoning, DNS poisoning still represents a viable threat within today's environment, though its effectiveness has been reduced in certain conditions.

HTTPS (Hypertext Transfer Protocol Secure) utilises SSL/TLS encryption as well as digital certificates to verify a website's identity (Ciampa, 2024). An attacker can poison the DNS entry of a victim and redirect them to a false domain. However, HTTPS also means that attackers' servers are unable to present a valid SSL certificate for the target domain. Therefore, when a browser sees this it will be able to identify the problem and show the user a warning stating that there is a security issue with the site they were attempting to reach (Cloudflare, n.d.). This is a major mitigation: Given that many large banking, government and commercial sites now utilise HTTPS, any poisoning attacks that redirect victims to a false version of these sites using invalid certificates will be immediately recognisable by the users' browsers through the display of visual warning messages which may alert the users to potential issues prior to any damage being done (Ciampa, 2024).

However, DNS poisoning has never been eliminated. Several vulnerabilities create potential openings for DNS poisoning, the main one being that the design of DNS was created for a much less complex internet and was based on trusting others (Cloudflare, n.d.). Therefore, all sites which are currently utilising only HTTP will be most definitely susceptible to DNS poisoning. In addition, attackers may acquire free SSL certificates through services such as Let's Encrypt in order to create fake webpages with convincing similarity to legitimate ones, for example `www-fnb.co.za` as opposed to `www.fnb.co.za`. This creates the possibility of using the same techniques as DNS poisoning even when using HTTPS by simply requiring users to pay close attention to URL details. Browsers mainly display security warnings when HTTPS certificate validation fails. The best technical method to mitigate DNS poisoning is by using DNSSEC (DNS Security Extensions), which adds a digital signature to DNS records allowing resolvers to verify the authenticity of their responses. However, global adoption of DNSSEC is currently incomplete (Ciampa, 2024; ISC, 2013).

To summarise, although HTTPS introduces an additional layer of protection against DNS poisoning through the utilisation of certificate verification, it does not eliminate this type of threat. Therefore, as long as there are sites that utilise only HTTP, users ignore browser warnings regarding security risks, and global adoption of DNSSEC is not complete, DNS poisoning will continue to be a viable attack vector in today's cybersecurity landscape.

Question 4

1. The Role of PKI in Cryptography

A Public Key Infrastructure (PKI) is a networked collection of processes, protocols, hardware, and software used to issue, manage, store, revoke, and distribute digital certificates along with managing public-key encryption for enabling secure communications over the internet and other networks (IBM, n.d.). PKI has been particularly valuable for providing secure communication capabilities for various applications such as internet banking, e-commerce, and secure e-mail (IBM, n.d.).

Asymmetric cryptography forms the basis for PKI. Asymmetric cryptography is a method of encrypting messages using two mathematically related keys; one public key, which may be widely distributed, and a private key that is maintained by the holder of the public key (Ciampa, 2024). Messages encrypted with the public key can only be decrypted with the corresponding private key. Therefore, only the intended recipient will have access to the message.

The PKI model works by associating a digital certificate with a verified identity. Digital certificates serve as an electronic identification document proving the identity of a user, organisation, or system. The primary elements of a PKI include a CA (certificate authority), RA (registration authority), digital certificates, and a public-private key pair that performs the encryption and decryption operations (IBM, n.d.; Ciampa, 2024).

PKI provides four major security features: confidentiality through encryption; authentication for verifying identity; integrity for preventing alteration of data; and non-repudiation for preventing users from denying their action (Ciampa, 2024; SSL.com, 2024). PKI establishes the foundation necessary to provide secure communication capabilities across digital media.

2. Importance of PKI in the South African Banking Sector

In addition to its use in many other industries, PKI also has an important function within the South African banking sector. In this industry, most financial transactions occur via the internet. Due to the volume of transactions occurring via the internet and strict regulatory guidelines, PKI is required for banks to securely facilitate online transactions while protecting customers' personal identifiable information and guaranteeing that users are accessing legitimate systems.

In practice, public key infrastructure (PKI) is applied in the online banking industry through mobile banking apps, ATMs and cards and trusted communications between banks. For example, when you log on to online banking from a reputable bank like Standard Bank, Absa, First National Bank or Capitec, PKI will be utilised to create an encrypted link between your device and the bank's servers and validate whether the site is legitimate. A pad lock symbol displayed by your browser verifies that a valid SSL / TLS certificate is being utilised which was issued through a process using PKI (Entrust, n.d.; ITWeb, 2023).

As mentioned previously, PKI plays a vital role in the online banking industry by protecting against fraud, identity theft, securing sensitive financial information, ensuring secure authentication and enabling the application of digital signatures for transactions. Digital signatures enable non-repudiation where both the customer and the bank cannot dispute having completed a transaction once it has been electronically signed. Electronic Signature Frameworks have provided legal recognition of digital signatures in South Africa (Ciampa, 2024; DPSA, 2019).

Additionally, PKI assists organisations in meeting their security obligations under POPIA (POPIA, 2013), which stipulates that all organisations need to apply reasonable measures to ensure that they protect individuals' personal data. Online banking platforms without PKI would be at significant risk of fraudulent activity including phishing attacks on customers and fake sites created to obtain access to customers' login credentials as well as data breaches.

3. PKI Trust Models

Trust Models describe how trust is established amongst users, systems and Certificate Authorities within a PKI Environment (Ciampa, 2024). There are three main trust models: Hierarchical Model, Distributed Model and Bridge Model.

A. Hierarchical Trust Model

The hierarchical trust model is the most commonly used PKI model and is modelled similarly to a tree. The hierarchy starts at the "Root CA" (Certificate Authority), which is located at the top. Beneath the "root", there are multiple levels of "Subordinate CAs". Each level represents a level of trust. Trust flows from the root CA down to all other entities

In the South African banking context, this model is used in HTTPS connections. Banks obtain SSL/TLS certificates from trusted Certificate Authorities, and when a user visits a banking website, their browser verifies the authenticity of the certificate through a trusted Certificate Authority to confirm the site is legitimate (ITWeb, 2023; Ciampa, 2024).

The advantages of the Hierarchical Trust Model include simplicity and ease of management. It is easy to determine who has authority over whom, because of the clear "chain of trust". On the other hand, one major drawback of the Hierarchical Trust Model is reliance on a single entity: the root CA. For example, if the root CA is compromised, then the entire trust structure is compromised (Ciampa, 2024).

B. Distributed Trust Model (Web of Trust)

Unlike a centralised trust model, which uses a root CA as its primary component for establishing trust among users, a distributed trust model, also referred to as a Web of Trust, uses multiple entities that have verified and signed each other's public keys to create a network of interdependent trust relationships. This means that no one single entity can compromise the entire system (Axelspire, 2026; Ciampa, 2024).

In the South African banking sector, this model is less suitable for customer-facing systems in highly regulated environments such as banking, where centralised control is

required. Although this model could be used in some internal contexts within organisations where different departments or systems require trust establishment without a single source of authority (Ciampa, 2024).

An Advantage of using a distributed trust model includes reducing the number of single points of failure to improve resiliency; however, the disadvantage includes difficulty in managing this type of system across large environments with strict regulations such as banking (Axelspire, 2026; Ciampa, 2024).

C. Bridge Trust Model

The Bridge Trust Model links many independent PKI systems via a Bridge Certificate Authority. Therefore, when organisations need to have trust in one another's communications, they do not need to use the same Root CA (Ciampa, 2024).

In the South African banking sector, this model has a significant value in scenarios where secure interoperability is required amongst the various financial institutions within the banking sector. For example, when using systems like the National Payment System run by the South African Reserve Bank, there needs to be secure communication between Banks. A bridge trust model could be utilised to enable two separately operated PKI systems to place confidence in each other's communications when conducting an interbank transaction (SARB, n.d.). According to the Literature on PKI trust models, a Bridge CA can connect different hierarchical architectures and allow different organisations to operate their own PKI structures whilst maintaining secure communication with one another via a single trusted connection point (Axelspire, 2026; Ciampa, 2024).

One of the primary advantages of implementing a Bridge Trust Model is that it enables secure interoperability between organisations. However, it is typically much more difficult to implement and manage than the hierarchical and distributed models. Additionally, should the Bridge CA be compromised, it could potentially affect multiple related systems (Ciampa, 2024).

4. Conclusion

PKI forms a fundamental component of modern cryptography and is essential for establishing trust in digital communication systems. In addition to providing confidentiality, authentication, integrity and non-repudiation it is also necessary to protect sensitive data and financial transactions.

Each of the three models provide a different type of security. The Hierarchical Model is suitable for everyday banking transactions as it provides a well-established and widely accepted framework. The Distributed Model is best suited for small internal systems or where flexibility is needed. The Bridge Model enables secure communication between different banks and institutions.

Overall, PKI and its trust models form the basis of trust, security and reliability in South Africa's digital banking environment.

Question 5

1. Optimal Installation Practices:

The installation of WLAN devices and how these are configured is a key first line of defence against cyber threats to small, medium and micro enterprises (SMMEs) in South Africa. This is due to the fact that most businesses do not have large IT budgets or dedicated personnel responsible for securing their networks; further, many will utilise routers supplied by Internet Service Providers (ISPs), which typically have a default configuration setting (ZenDefend, 2026).

Access Point Placement

A wireless Access Point (AP) is the device that broadcasts the Wi-Fi signal across a premises. Where a Wireless Access Point (AP) is placed has a significant impact on both the quality of coverage and the level of security that can be achieved (Ciampa, 2024). The typical broadcast range of an indoor Access Point is 150 to 300 feet, whilst outdoor signals can be much stronger, extending up to 1,000 feet. Therefore, if a business fails to properly secure its wireless network, then it risks having its Internet connection available to anyone who wishes to connect to it. As a result, for South African SMMEs operating in shared commercial environments such as shopping malls or office parks, there is a very real threat that the network of one business could potentially be detected by neighbours or those accessing the car park area around the premises (CISA, 2021).

Therefore, best practice dictates that APs are installed centrally within a building with the least possible number of access points located near windows or exterior walls. In doing so, you concentrate your Wi-Fi signal inside the building and minimise any potential leakage of the signal outside (Ciampa, 2024). Additionally, APs should be installed in locations that make them difficult to interfere with, all cabling associated with the network should be properly secured and wiring closets should be locked and monitored (ITU Online, 2026). On larger premises it is generally recommended that you deploy multiple APs set to operate at lower powers than would be required for a single high-powered AP. In doing so you reduce the risk of Wi-Fi bleed occurring beyond the business boundaries while still providing good quality coverage throughout the business premises.

Signal Strength Settings

Wireless signals have an impact on how far they travel. From a security point of view, the signals coming out of your access points (APs) need to be kept as weak as possible. In general, you would want to conduct a site survey with various values to find what value works best for your needs to ensure you have coverage in all the areas used by employees/users. It's also important to keep in mind, that you do not want to allow your signal to extend into public areas such as parking lots, streets, etc., which would enable unauthorised individuals to gain access to your wireless network (Cisco, 2023; Ciampa, 2024). A practical goal for a South African SMMEs, would be to have enough coverage throughout their working space, while limiting the signal from extending into public areas. Maintaining this balance between adequate coverage and controlling the amount of signal radiating from the building is a key aspect of secure wireless practices

(Ciampa, 2024). If you do not limit the amount of signal radiated from your premise, there are higher chances of unwanted/unauthorised connections being made to your network via an outsider accessing your network. This highlights that signal strength directly affects both network coverage and the likelihood of unauthorised access.

2. Configuration Options:

Spectrum Selection

Modern Access Points (APs) supports multiple frequencies. Selecting the right frequency when using wireless technology can have significant security consequences. The 2.4 GHz frequency provides better distance than the other two frequency options but also allows for greater penetration through walls/open spaces and provides a higher likelihood of having your signal extend past the boundary of your business premise. The 5 GHz frequency option provides faster speeds, less interference, and has a smaller footprint, allowing it to stay confined inside the perimeter of your business premise (Ciampa, 2024; Intel, n.d.). Since channel overlap is limited in 2.4 GHz usage, most businesses opt to use less 2.4 GHz and rely more on 5 GHz and/or 6 GHz, when available, for business operations (ITU Online, 2026). As a result, South African SMMEs should prioritise the 5 GHz frequency for all their business operations and reserve 2.4 GHz only if there is a legitimate reason to require additional range and cannot achieve the desired coverage requirements utilising 5 GHz or another supported frequency. The choice of frequency directly influences how far the network signal extends and the potential exposure to external threats.

Antenna Placement

The type and location of antenna's affectively impact on what direction a wireless signal will go. Omnidirectional antennas send out a wireless signal in every direction. The result is that you may have some signal leaking through your exterior walls. Directional antennas are designed to point the signal inside your building. As a result, the signal is concentrated in one area and reduces the amount of signal that leaks out into the surrounding area (Ciampa, 2024). Effective RF Planning results in the most efficient possible position for each access point as this reduces dead zones and limits the amount of potential interference that could be used for malicious purposes. Channel allocation allows you to prevent overlapping channels that can be used to create malicious interference or eavesdrop. Well planned layouts also limit how easy it is for unauthorised users to connect to your network from an outside source (ITU Online, 2026). For example, a South African SMME operating in an open plan environment would benefit from having their Access Points mounted on the ceiling with downward facing antennas. These antennas help concentrate signal throughout the floor space as opposed to being directed outward toward neighbouring business. This will allow them to reduce the exposure of their wireless signals to unauthorised users.

Additional Configuration Options

There are numerous configuration options that need to be considered when it comes to securing a Small Medium Enterprise's (SMME) wireless network beyond the initial considerations of placement and spectrum. Currently, WPA3 is the highest level of encryption available today. It is recommended that if you have equipment that does not

support WPA3 to update your equipment as using either WPA or WPA2 may potentially expose your network to hackers (CISA, 2021). In addition to encrypting your wireless network, most wireless routers include an option to hide the SSID. Hiding the SSID will make it harder for hackers to find your wireless network. At the minimum, it would be wise to change the SSID name to something unique rather than having it set to the original SSID name provided by the manufactures. If you don't change the SSID name, this may provide enough information to a hacker on what type of router you are running (CISA, 2021). As previously mentioned, changing the default administrator password is crucial to protecting your wireless network. The default administrator password can be found online and is one of the most commonly used passwords by hackers when attempting to gain access into a small business's network (CISA, 2021) Establishing a separate guest Wi-Fi network from your main network protects your organisational information from being accessed through your guest Wi-Fi network. This is particularly important for SMMEs based in South Africa that offer customer Wi-Fi services. By creating a separate guest Wi-Fi network, you prevent customers' devices from accessing any internal business systems (CISA, 2021).

Each of the above-mentioned methods help comply with South African's Protection of Personal Information Act (POPIA, 2013), which require organisations to take adequate technical measures to protect individuals' personal information. If your organisation has a poorly secured WLAN that results in a data breach, the maximum penalty amount is R10 million (ZenDefend, 2026; POPIA, 2013).

Relevance to South African SMMEs

The significance of wireless security is especially relevant to South African SMME's. Many businesses utilise wireless networks as their primary method of accessing the Internet, however, due to limited resources they may have limited or no advanced cybersecurity measures available to them. If an organisation does not properly configure its WLAN it could be compromised resulting in theft of customer data, financial fraud and potentially reputational damage. (GP TechSupport, 2025)

If a business implements the correct methods for installing and configuring their WLAN it will allow them to protect sensitive customer data and assist organisations in complying with The Protection Of Personal Information Act that has been put into place requiring all organisations to protect personal data. (SME South Africa, 2024).

Conclusion

Therefore, securing your Wireless Local Area Network (WLAN) requires a combination of good physical installation and correct configuration. Properly placing Access Points, along with controlling signal strength, will help prevent unauthorised outside access. In addition to this, utilising configuration options including choosing a suitable spectrum, locating antennas correctly and ensuring you have sufficient security configured will provide an additional layer of protection. These steps will allow South African SMME's to operate securely, protect sensitive information and build trust with their customers.

Question 6

Dear Fellow South Africans,

There have been many issues related to energy across South Africa as a result of how we generate, distribute, sell and consume energy. One of the ways that we are working to address these issues is through the installation of smart electric meters. Smart meters can provide numerous benefits to customers by providing real-time pricing information and helping utility companies identify outages. However, there have been community-based objections to the implementation of smart meters. The purpose of this letter is to understand the objections raised against smart meters, to respond to those objections based upon existing research and to discuss why I believe that smart meters are better than analogue meters.

1. Investigating Resistance to Smart Meters

It should come as no surprise that many South Africans have expressed hesitance regarding the acceptance of smart meters. Research examining consumer perceptions of smart meters among South African consumers revealed five primary drivers of that resistance including a lack of trust in the technology, concerns about monetary cost, and uncertainty around ease of use (Muchenje & Botha, 2021). These factors directly influence whether consumers are willing to accept and use smart meter technology.

Privacy is another major concern. Concerns exist among the public that smart meters may capture extensive details concerning their electrical consumption patterns, which may then be used improperly or without the consumer's consent (Mobility Foresights, 2025). Historically, South Africans have experienced problems with their municipal services, such as incorrect billing and poor service delivery, thus, creating distrust among some citizens toward both municipalities and Eskom. Furthermore, some residents are concerned that smart meters will allow utility providers to disconnect their electricity supply remotely. Lastly, although progress has been made since the advent of mobile devices in terms of bridging the "digital divide," a large percentage of South Africans still do not possess sufficient digital knowledge and/or broadband connectivity necessary to fully participate in smart-meter platforms (ITWeb, 2024).

2. Addressing These Concerns

While many of the above-mentioned concerns are legitimate, a large amount of the opposition to smart meters has stemmed from a lack of knowledge of how smart meters really function. Smart meters were created to give both utility providers and consumers real time and accurate consumption data. The data collected is not used for personal surveillance, but rather for improving billing accuracy and operational efficiency. Improved billing accuracy and operational efficiency protects consumers from the over-charging that has historically resulted from estimated readings on analogue meter systems. (Conradie, 2024)

Smart meters support demand response programs which assist utilities in balancing supply and demand more effectively. Therefore, better grid management reduces the likelihood and duration of outages. Additionally, remote monitoring capabilities allow faults to be detected

and resolved faster than traditionally possible, improving service delivery; something South Africans have been calling for, for a long time. (Conradie, 2024).

Regarding privacy, it is important to note that smart meter data is utilised for operational purposes and not personal surveillance. In addition, with South Africa's data protection laws under the Protection of Personal Information Act (POPIA), utility providers are legally obligated to handle consumer data responsibly (ITWeb, 2024). Concerning cost, smart meters can help protect consumers financially by eliminating billing errors caused by inaccurate manual estimates.

3. Benefits of Smart Meters Over Analogue Meters

Compared to traditional analogue meters that rely on manual readings & provide limited feedback to users, smart meters offer real-time data, automation, & significantly improved accuracy (Conradie, 2024).

The first and most immediate advantage is accurate billing. Analogues do not have the ability to deliver precise, real-time usage data like smart meters do. Consequently, analogues often produce estimated bills based upon human error during manual readings. Conversely, smart meters deliver exact, real-time usage data ensuring consumers pay only for what they actually use (Conradie, 2024; ITWeb, 2024).

Secondly, smart meters enable real-time energy monitoring. This allows households and businesses to see how much electricity they are using at any given time. By being able to track their usage in real-time, consumers can adjust their behaviour to use less power; identify wasteful appliances that use more than needed; and therefore, reduce their electricity bills. Additionally, by enabling consumers to know where they stand with respect to their energy usage, it promotes a greater sense of responsibility and energy awareness. (Conradie, 2024).

Thirdly, from a grid management point of view, smart meters provide utilities with vital data which will allow them to manage demand for electricity better than previously possible. By having this improved visibility into electricity demand, utilities should experience fewer and shorter power outages which is another benefit of installing these devices in a country like South Africa where load shedding still has daily life implications. (Conradie, 2024).

Fourthly, smart meters contribute to environmental sustainability. By encouraging efficient use of electricity and supporting renewable sources of energy, they help reduce all overall energy waste. This is consistent with goals set by government for sustainability in South Africa. (Conradie, 2024).

Finally, by removing the need for utility staff to take physical meter readings, smart meters also help save costs on operations as well as eliminate the chance of errors when collecting data (Conradie, 2024). This could result in improved service delivery and ultimately, better resource allocation by municipalities.

4. Conclusion

Smart meters are an important step forward for South Africa's energy industry. Although resistance to these devices exists within many communities as well as from individuals in those communities; this resistance is primarily based upon misunderstandings regarding

what smart meters do and how they work. There has been considerable research into the benefits of smart meters and there is evidence showing that smart meters will result in more accurate billing, better energy management, increased reliability/stability of the electrical distribution system and ultimately better opportunities for sustainable practices. Therefore, smart meters not only modernise how electricity is managed in South Africa, they also contribute significantly to greater levels of transparency, efficiency and sustainability within South Africa's energy industry.

Yours sincerely,
Kandyce Jade Smit

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